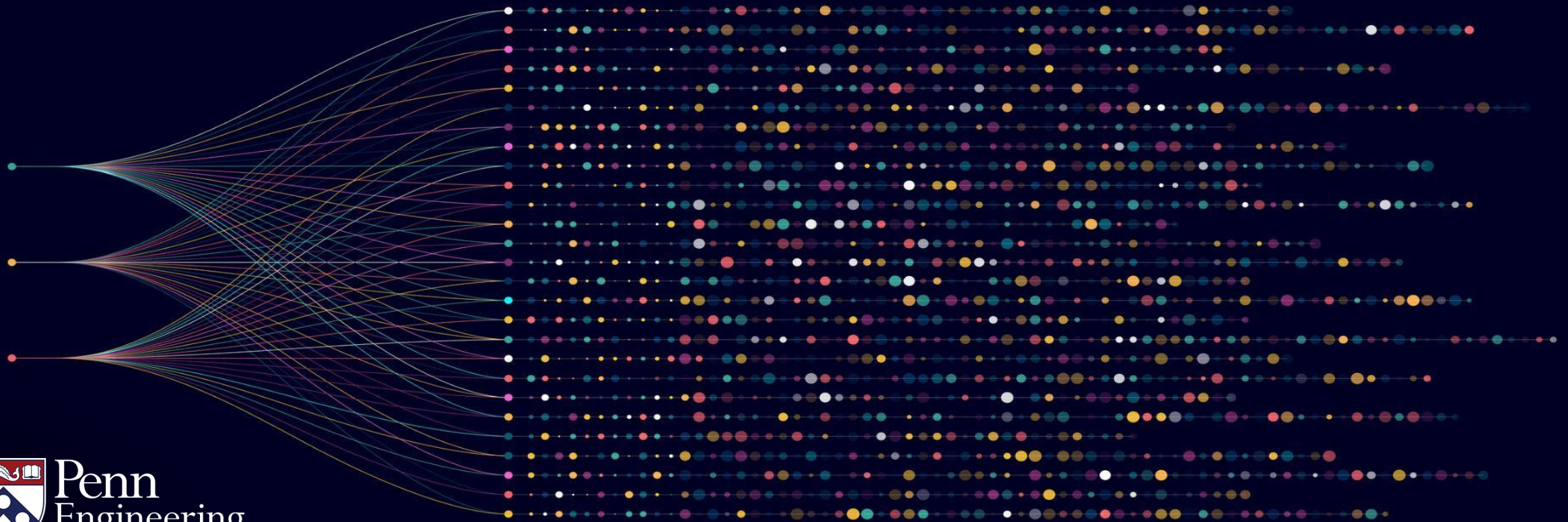


Algorithms for Big Data

Anindya De

Erik Waingarten

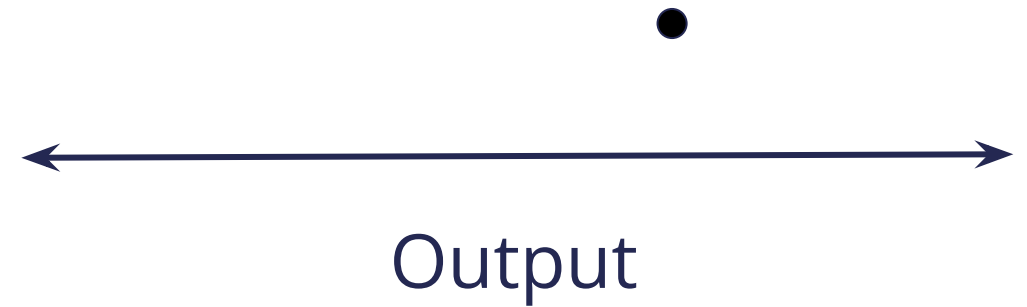
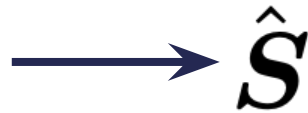
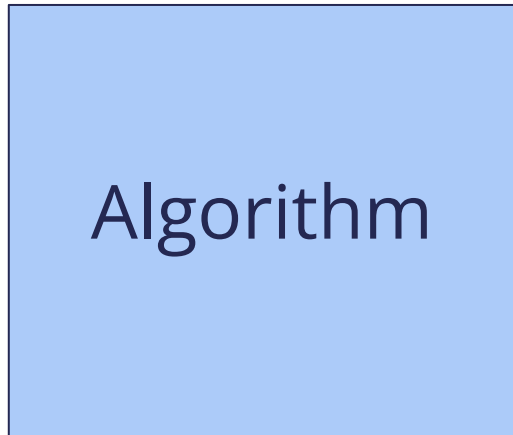
Randomization: Pros and Cons



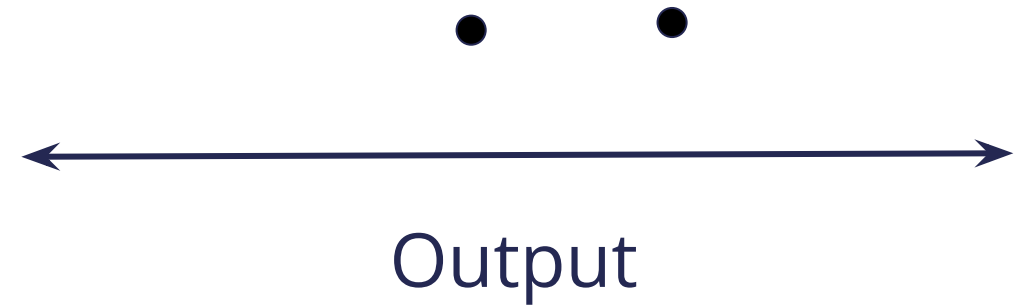
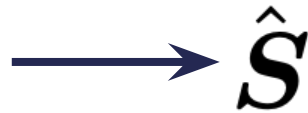
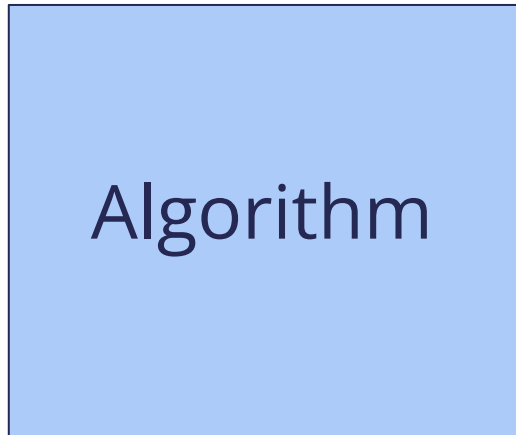
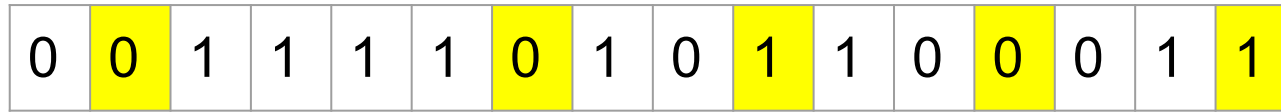
Objectives: Computing Large Averages

- Randomized algorithms
- Pros and cons of randomization

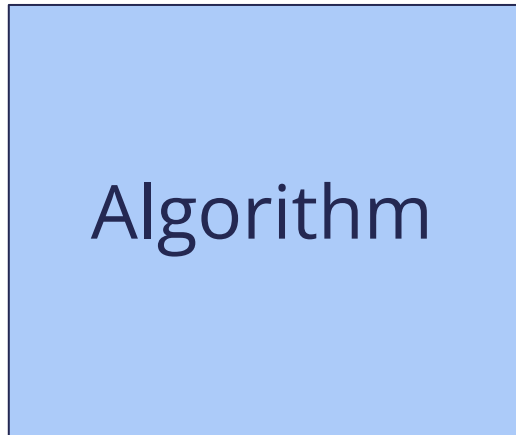
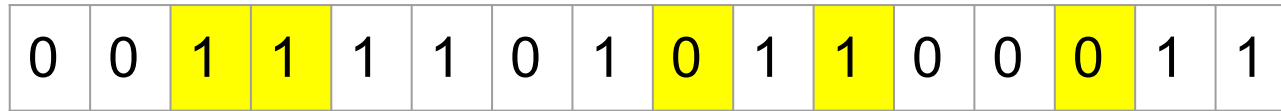
What if the algorithm does not commit to certain queries?



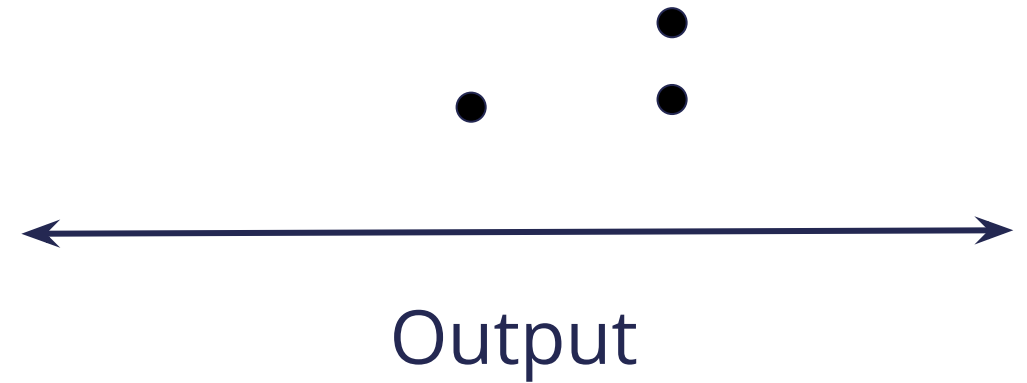
What if the algorithm does not commit to certain queries?



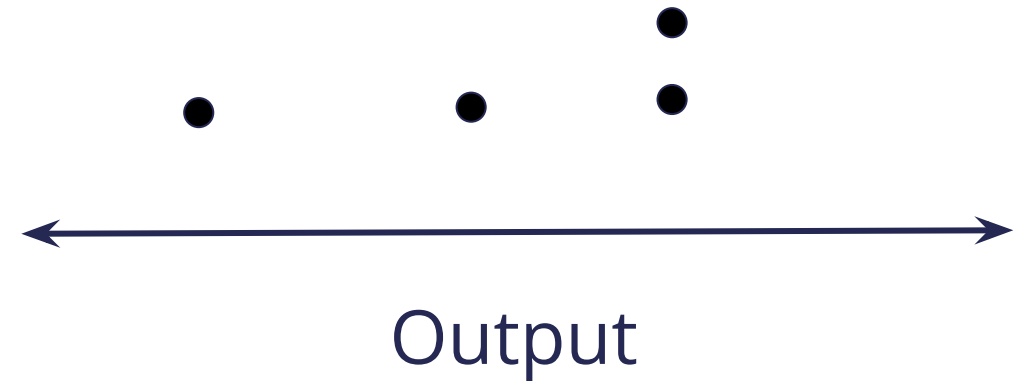
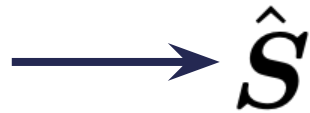
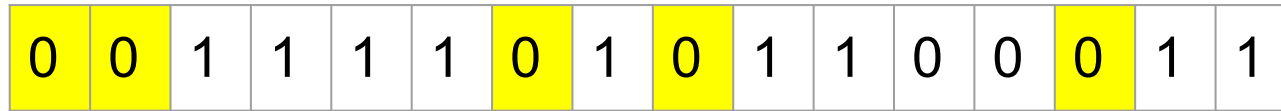
What if the algorithm does not commit to certain queries?



→ $\hat{S}$

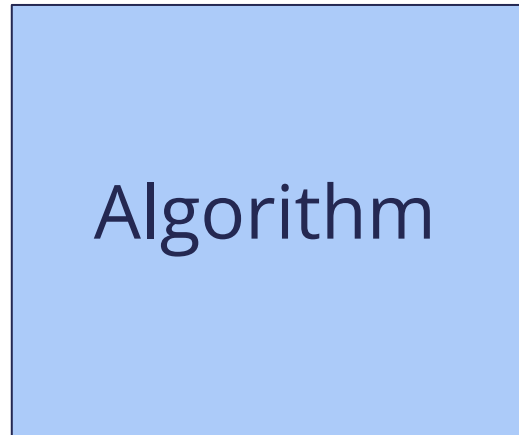


What if the algorithm does not commit to certain queries?

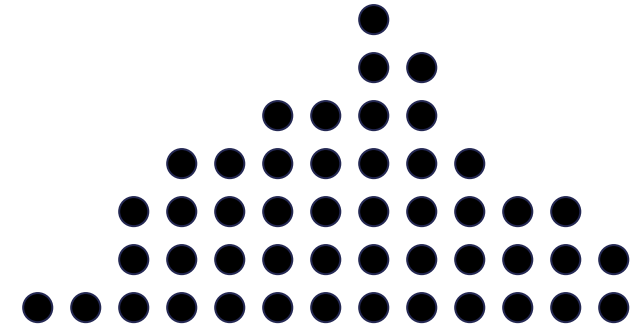


What if the algorithm does not commit to certain queries?

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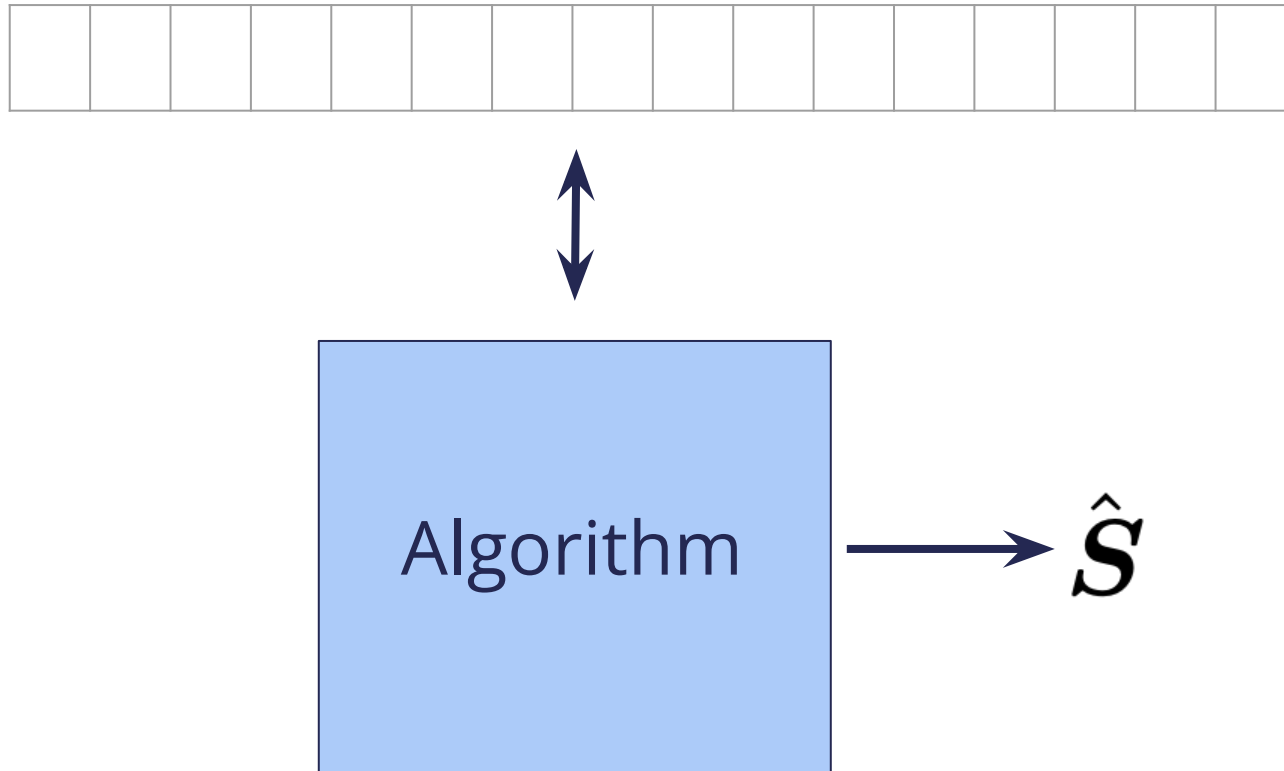


→ $\hat{S}$



Output

Visualizing a randomized algorithm



- Output of the algorithm is a random variable
- Correctness in terms of properties of the random variable

Pros and Cons of Randomized Algorithms

- Pros:

- Algorithm doesn't commit to fixed set of queries
- Cannot construct indistinguishable instances, because we don't know queries

- Cons:

- Every time we execute algorithm, may get a different answer
- Hard to debug
- Algorithm's output is a random variable, must reason probabilistically about guarantees